

RESEARCH ARTICLE

Market power in the German dairy value chain

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Abstract

In this paper, the NEIO approach is extended to allow for oligopsony power in successive markets of a value chain. Two price equations are deduced from simultaneous partial equilibria of the endogenous variables and are embedded in a VECM to account for a long-run cointegration relationship. The model is estimated via the Kalman-Filter to allow for time variation in the long-run parameters, and a dynamic factor model used to extract a common factor from the time-variant coefficients. The results are then used to calculate the industry average input conjectural elasticities. The framework is applied to German dairy value chain data over the time period of January 2000 to March 2011. The results indicate almost perfect competition in the raw milk market and imperfect competition in the dairy output market, which is however far from monopsony. [EconLit citations: L13, D43, C32, Q13].

1 | INTRODUCTION

For a long time, agricultural and food product markets were regarded as prime examples of perfect competition (Sexton, 2013). This notion started to erode in the first few decades after World War II. Heavy concentration and consolidation at the food processing level remodeled the structure of most agri-food value chains into the shape of an hourglass with many atomistic farmers at the bottom, a few larger food processors in the middle, and a vast number of heterogeneous distributors, for example wholesalers and retailers, at the top (Sexton, 2000). Simultaneously, bargaining power was shifted within these chains in favor of few food processors, because the retail sector maintained its heterogeneous structure and the farm sector its atomistic one. For example, 50% of the value added in the U.S. food processing sector was achieved by only 20 manufacturers in 1995, whereas the share was only half this size in 1954 (Sexton, 2000). Consequently, economic research in the 1970s and 1980s focused on the food processing sector as a possible source of oligopoly power with negative welfare implications for retailers and final consumers. The belief of farm commodity markets being perfectly competitive persisted.

In the late 1980s, the focus slowly shifted towards the idea of oligopsony power in raw agricultural commodity markets, as the atomistic farm sector interacted with highly concentrated food processing industries (e.g. Muth & Wohlgenant, 1999; Schroeter, 1988; etc.). Nevertheless, research on oligopsonistic behavior of food processors has remained scarce, for two main reasons. First, most empirical applications that are capable of evaluating the extent of market power are mainly based on the new empirical industrial organization (NEIO) framework, in which quantity data on all inputs are required for the estimation. In many cases, these data are scarcely available to researchers

(Muth & Wohlgenant, 1999). Second, even though the high level of oligopsony power can have devastating effects on farmers, the farm share in the value of food purchases prices is less than 1/6, e.g. 15.8% in the United States in 2008 (Canning, 2011). Consequently, oligopsony power in food industries had never been of large interest to researchers as the main source of welfare losses (Sexton, 2000).

Recent developments in food retailing seem to have changed the bargaining power along the agri-food value chains. Starting from the mid-1990s, mergers and acquisitions have resulted in higher levels of retail concentration, much higher than that of the food processing industry in the United States and European Union (EU). In the EU, for example at the beginning of the 2000s, the top 20 retailers accounted for 40% of the total industry revenues, in contrast, the top 20 food processor only accounted for 15% (Clarke, Davies, Dobson, & Waterson, 2003). By the mid-2000s, in some EU countries, such as Germany or the United Kingdom, the five largest enterprises made up more than 70% of retailers' total turnovers (CCMI, 2008). In addition to the growing concentration, retailers have integrated wholesaling into their business and further taking control over the upstream stages by demanding specific products and creating own brands (Dobson, Waterson, & Davies, 2003). Consequently, manufacturers were replaced by retailers as the drivers of agri-food value chains (Bell, Davies, & Howard, 1997).

Since the 2000s, the focus of market power research has shifted from the processing sector to the retail sector. In general, most applications concentrated on the effects of retail concentration on consumers' welfare. Nevo (1998) developed a framework to measure the extent of market power in differentiated product markets. Building on this model, Nevo (2001) investigated the U.S. cereal industry by predicting the price-cost margins and comparing them to price-cost margins of competing models. This model separates the extent of market power (price over marginal costs) by retailers in three different components: product differentiation, portfolio effect, and price collusion. Empirical applications to various product groups mainly found that the markups were not due to price collusion but rather results of a combination of product differentiation and the portfolio effect (i.e. Nevo, 2001; Rojas, 2008; Slade, 2004).

Another branch of economic literature acknowledged the high levels of concentration at the food processing stage. Villas-Boas and Hellerstein (2006) developed a structural model of successive oligopoly and proved that the oligopoly-pricing behavior is identified in homogenous and differentiated product markets assuming a fixed proportion technology, nonconstant markups, and both nonlinear and linear demand relationships. Cakir and Balagtas (2012) adopted this model to investigate the extent of oligopoly power along the U.S. dairy supply chain in the presence of dairy cooperatives. They found no evidence of noncompetitive behavior in most U.S. fluid milk marketing regions, neither at the food processing nor the retail level (Cakir & Balagtas, 2012).

Although previous research mostly concentrated on the issues of oligopoly power, research focus, in particular in Europe, has recently shifted towards studying retailers' purchasing power on the food processing industry. Gohin and Guyomard (2000) were the first to analyze the French retail sector for oligopsony power, but many other studies (i.e. Fałkowski, 2010; Lloyd, McCorriston, Morgan, Rayner, & Weldegebrüel, 2009; Madau, Furesi, & Pulina, 2016) have followed. The growing research interest in retailers' oligopsony power has been triggered after the food price peaks in 2007/2008, during which consumer food prices and agricultural producer prices experienced similar trends of rapid growth, after which producer prices quickly dropped below the pre-2007 level in 2009 while consumer prices remained high. The European Commission (EC) expressed concern over the developments and warned of negative long-term effects of retailers' oligopsony power for the entire agri-food sector (EC, 2009).

Even though research interest in retailers' oligopsony power sparked after 2009, to date, there have only been three studies to the best of our knowledge that have estimated the actual extent of retailers' oligopsony power in agri-food markets. These are the previously mentioned Gohin and Guyomard (2000) for the French retail sector, Anders (2008) for the German, and Salhofer, Tribl, and Sinabell (2012) for the Austrian. Furthermore, so far, no model exists that simultaneously accounts for the threat of buyers' power of retailers over processors and of processors over farmers. Therefore, the aim of this study is to develop a model that allows for identifying oligopsony power in the subsequent markets of an agri-food value chain. The approach is applied to the German dairy value chain, because the German Federal Cartel Office (FCO) has recently investigated the domestic raw milk as well as dairy product markets for oligopsonistic conduct (BKartA, 2012). Because the final report of the antitrust agency acknowledged that the structure of the German dairy value chain is open to the abuse of market power in its successive markets but failed to provide any evidence

for oligopsony power, another objective of this study is to empirically investigate the German dairy value chain for the presence of oligopsony power. The time period of the analysis spans from January 2000 to March 2011, a period characterized by heavy structural changes and important policy reforms, e.g. the gradually increasing milk quota since 2006.

The theoretic structural model of successive oligopsonies is laid out in Section 2. In Section 3, two price equations including exogenous cost shifters are derived from the structural model. Using only two equations significantly lowers the data requirement compared to other NEIO empirical applications, since, to estimate market power parameters, both in the upstream and downstream market, neither input nor output quantity data are required. Section 4 deals with the structure of the German dairy value chain and the database. The empirical results, as well as the evidence of market power and its magnitude in the German raw milk and dairy products markets are presented and discussed in Sections 5 and 6, respectively. Our paper ends with concluding remarks that comprise limitations of the approach and implications for future research.

2 | THEORETICAL MODEL

The theoretical model developed here is based on that of Muth and Wohlgenant (1999), extended to allow for an oligopsonistic structure of the downstream market. To model technology at all stages, we use second order-differential quadratic forms for all production, cost, and revenue functions (for the explicit functional forms see the Appendix; Chambers, 1988). The upstream market comprises farmers as the exclusive supplier of a specialized good, and processors as the exclusive buyer. Given the farmers' cost function and the assumption that farmers procure the necessary inputs for their production process in competitive markets, the inverse supply equation of the specialized good can be denoted by

$$p^f = S^{f-1}(x^f, z_h^f, t^f), \quad (1)$$

where z_h^f is a h vector of input prices, x^f is the aggregated supply quantity, and t^f is a trend variable depicting technical change at the farm level f . Given the supply relation and the assumption that n processors procure and process the farm output as well as sell the processed product as price takers, we can formulate the i th processor's profits as

$$\pi_i^p = R_i^p(p^p, x_i^f, x_{ki}^{z*}, t^p) - p^f x_i^f - z_{ki}^p x_{ki}^{z*}, \quad (2)$$

where p^p is the output price, $R_i^p(\bullet)$ is the revenue function of the i th processor, $x_{ki}^{z*} = x_{ki}^z(x_i^f, p^p, z_{ki}^p)$ and z_{ki}^p are quantity and price vectors of k nonspecialized inputs, and t^p captures technological change at this stage. The quantities of the nonspecialized inputs are assumed to be at their optimal level, conditional on the i th processor's chosen level of the specialized input x_i^f (Muth & Wohlgenant, 1999). The first-order condition with respect to the i th processor's choosing the quantity of x_i^f is

$$\frac{\partial \pi_i^p}{\partial x_i^f} = \frac{\partial R_i^p(p^p, x_i^f, x_{ki}^{z*}, t^p)}{\partial x_i^f} + \frac{\partial R_i^p(p^p, x_i^f, x_{ki}^{z*}, t^p)}{\partial x_{ki}^{z*}} \frac{\partial x_{ki}^{z*}}{\partial x_i^f} - p^f - \Theta_i^p \frac{\partial S^{f-1}(x_i^f, z_h^f, t^f)}{\partial x_i^f} x_i^f - z_{ki}^p \frac{\partial x_{ki}^{z*}}{\partial x_i^f} = 0. \quad (3)$$

Under the assumption that the nonspecialized inputs x_{ki}^{z*} are procured in competitive input markets, the terms of equation (3) can be rearranged to yield

$$p^f \left(1 + \frac{\Theta_i^p}{\varepsilon^f} \right) = \frac{\partial R_i^p(p^p, x_i^f, x_{ki}^{z*}(x_i^f, p^p, z_{ki}^p), t^p)}{\partial x_i^f}, \quad (4)$$

where the observed price of the specialized input plus a relative markdown equals the marginal revenue product value for the specialized input. The term $\Theta_i^p = \frac{\partial x_i^f}{\partial x_i^f} \frac{x_i^f}{x_i^f}$ represents the input conjectural elasticity of the i th processor in the specialized input market, whereas $\varepsilon^f = \frac{\partial x^f}{\partial S^{f-1}} \frac{S^{f-1}}{x^f} = \frac{\partial x^f}{\partial p^f} \frac{p^f}{x^f}$ is the market price elasticity of input supply for the farm output (i.e. the specialized input). The input conjectural elasticity measures the responses in total industry input quantity to a

change in the i th processor's input level. If we aggregate across all firms, $\Theta^p = \frac{1}{n} \sum_{i=1}^n \frac{\partial x_i^f}{\partial x_i^p}$ becomes the average input conjectural elasticity (CE)¹ of the processing industry since we assume that aggregation is achieved by averaging over all processors' marginal product (Muth & Wohlgemant, 1999). The CE can take values from 0 to 1. A value of 0 would indicate that the market is competitive, while a value of 1 indicates monopsony, or a cartel that acts as a monopsony. Values that lie between 0 and 1 denote varying degrees of oligopsonistic market structures with the special case of the Cournot outcome at the value of $1/n$ (Bresnahan, 1989).

The downstream market for the processing industry output is modeled in a similar fashion as the upstream market. In the output market, processors are assumed to be price takers and to sell all their products to the retail sector. Applying Hotelling's Lemma to the processing industry's aggregated profit function² π^p provides us with the quantity of the processing industry's supply y^p to the downstream market:

$$\frac{\partial \pi^p(p^p, p^f, z_{ki}^p)}{\partial p^p} = y^p. \quad (5)$$

Accounting for imperfect competition in the form of oligopsony power (that is, p^p depends on y^p), the j th retailer's demand for the processors' output can be derived. Profit maximization behavior implies that the j th retailer's cost function is differentiated with respect to the level of the specialized input quantity yielding

$$\frac{\partial c_j^r(y_j^p, p^p, y_{mj}^{z*}(y_j^p, p^p, z_m^r, t^r))}{\partial y_j^p} = p^p + \Theta^r_j \frac{\partial S^{p-1}}{\partial y_j^p} y_j^p, \quad (6)$$

where the marginal factor cost of the output equals the price of the specialized input plus a relative markdown. The vectors z_m^r and t^r stand respectively for the nonspecialized input prices and the technological change at the retail level. The quantities of the m nonspecialized inputs are again assumed to be at their optimal quantity level $y_m^{z*} = y_m^z(y_j^p, p^p, z_m^r)$ conditional on the j th retailer's chosen level of the specialized input y_j^p . Aggregating across all retailers' cost functions by using a Gorman polar form, which allows firms to have individual but parallel cost functions so that marginal cost must equal across firms (Sperling, 2002), retailers' marginal costs can be formulated as

$$\frac{\partial c^r(y^p, p^p, y_m^{z*}(y^p, p^p, z_m^r, t^r))}{\partial y^p} = p^p \left(1 + \frac{\Theta^r}{\varepsilon^p} \right), \quad (7)$$

where $\Theta^r = \frac{1}{n} \sum_{j=1}^n \frac{\partial y_j^p}{\partial y^p} \frac{y_j^p}{y^p}$ is CE of the retail sector regarding the processor output product and $\varepsilon^p = \frac{\partial y^p}{\partial S^{p-1}} \frac{S^{p-1}}{y^p} = \frac{\partial y^p}{\partial p^p} \frac{p^p}{y^p}$ is the price elasticity of input supply for the processed product. The interpretation and the values of Θ^r are similar to those of Θ^p , and the ratio $\frac{\Theta^r}{\varepsilon^p}$ represents the retailers' ability to mark down the price of the specialized input.

Farm supply (1), processors' demand (4), processed good supply (5), and retailers' demand (7) form a system of equations. The system consists of four endogenous variables (p^p , p^f , x^f , and y^p), which can be directly influenced by the participants of the upstream and downstream market. Similar to Lloyd et al. (2009), we use a system of equations to find the explicit solutions of these endogenous variables. The result is that solely the exogenous variables³ will determine simultaneously the endogenous variables (for the detailed equations see in the Appendix (A6)).

In contrast to Lloyd et al. (2009) who subtracted the explicit solution of the upstream price from the one for the downstream price to derive a margin equation, we solve the explicit solution of p^p for both Θ^p and Θ^r . Inserting the

¹ The conjectural variations approach, the so-called "conduct parameter method" (CPM; Corts, 1999), has been criticized at various levels and its suitability for empirical application has been questioned. The main critique of this approach is that the static nature of CPM neither is able to capture nor to estimate the dynamic structure behind the firms' conjectures on the rivals' response to quantity changes (Friedmann, 1983; Corts, 1999). Furthermore, the CPM assumes that firms maximize profits in a single time period rather than a discounted stream of profits over a certain number of multiple time periods (Friedmann, 1983). However, Dockner (1992) showed that "a static conjectural variation analysis approximates long-run dynamic interactions" (Dockner, 1992: 390) and thus is suitable to contextualize dynamic conduct. Moreover, Genesove and Mullin (1998) were able to confirm the accuracy of the CPM concerning estimating conduct in an industry by using historical data on the US sugar industry, which included precise information on the marginal costs of individual firms. Consequently, we proceed with the CPM in the paper.

² $\pi^p(p^p, p^f, z_k^p, t^p) = p^p f(x^f(p^p, p^f), x_k^{z*}(x^f, p^f, z_k^p, t^p)) - p^f x^f(p^p, p^f) - z_k^p x_k^{z*}(x^f, p^f, z_k^p, t^p)$

³ Prices of all nonspecialized inputs and the technical change at all stages as well as the CEs of the processors and retailers.

obtained solutions for Θ^p and Θ^r into the explicit solution for p^f , yields the following two simplified linear price equations,

$$p^f = \beta^c(\Theta^p) + \beta^{p^p}(\Theta^p)p^p + \beta^{z_h^f}(\Theta^p)z_h^f + \beta^{z_k^p}(\Theta^p)z_k^p + \beta^{t^f}(\Theta^p)t^f + \beta^{t^p}(\Theta^p)t^p, \quad (8)$$

$$p^f = \beta^c(\Theta^r) + \beta^{p^p}(\Theta^r)p^p + \beta^{z_h^f}(\Theta^r)z_h^f + \beta^{z_k^p}(\Theta^r)z_k^p + \beta^{z_m^r}(\Theta^r)z_m^r + \beta^{t^f}(\Theta^r)t^f + \beta^{t^p}(\Theta^r)t^p + \beta^{t^r}(\Theta^r)t^r, \quad (9)$$

where the specialized input price is determined by the output price as well as the prices for nonspecialized inputs and technological change at all stages of the value chain. The β are parameters to be estimated and composed of the parameters of the previous system of four equations including the CEs (for details see Equations (A7) and (A9) in the Appendix). However, although Equation (8) is only determined by the variables and parameters of the supply and demand relations of the specialized input market, Equation (9) is comprised of the entire equation system, except for the CE of processors. Consequently, the β of Equations (8) and (9) further differ by solely consisting of one of the CEs,⁴ either of the processors or the retailers. The separation of the CEs to two separate price equations enables us to determine these independently.

As mentioned previously, the conjectural elasticities are dynamic concepts. Gollop and Roberts (1979) argued that conjectural variations (CV) ought to vary over time. The reason is that unlike technology, the CVs are directly formed and influenced by the firms' behavior through a procedure of adaptive expectations. Thus, we allow the firms' conjectures, and thus also the β parameters, to vary over time, as they are denoted in the rest of the paper by the subscript t .

Given the assumption of time-variation, the β parameters can be split into a time-invariant and a time-variant component (for the exact formulation see in the Appendix (A7) and (A9)),

$$\beta_t(\Theta_t) = \delta_o + \delta_\lambda \lambda_t(\Theta_t), \quad (10)$$

where the δ parameters are time-invariant and λ_t is a time-variant factor whose time variation is due to changes in the CEs. The estimation procedure allowing for the identification of the CEs is described in the following section.

3 | EMPIRICAL METHODS

Price time series, in particular of vertically or spatially integrated markets, often follow a common trend, resulting in nonstationarity of and cointegration among the time series (Hendry & Doornik, 2001). Variables are labeled as cointegrated when there exists one or more combinations of these variables being stationary in the long-run (Juselius, 2006). To account for the possibility of nonstationarity and cointegration, we embed our theoretical model in a vector autoregressive (VAR) framework. It is assumed that the data series of the endogenous covariates, p^f and p^p , can be approximated by the VAR representation,

$$p_t = \psi_1 p_{t-1} + \psi_2 p_{t-2} + \dots + \psi_p p_{t-m} + \Upsilon d_{t-1} + \varepsilon_t, \quad (11)$$

where p_t contains the two price series p^f and p^p , d_{t-1} comprises all k exogenous variables, and m is the number of lags. The size of k depends on which price equation is estimated. If Equation (8) is estimated, d_{t-1} contains all nonspecialized input prices at the farm and processing level, a constant and a time trend to capture technological change. For Equation (9) also the nonspecialized input prices at the retail level are added. The parameter matrices ψ_i ($i = 1, \dots, m$) and Υ are estimated by using a time series database ($t = 1, \dots, T$). The error term ε_t is a time-dependent vector of normal and identical distributed disturbances with zero mean and non-diagonal covariance matrix Σ .

⁴ Denoted by either Θ^p or Θ^r in the brackets behind the β parameters. If a β parameter is not followed by a bracket, it does not consist of any CE.

In the presence of cointegration relationships among the endogenous variables, (11) is rearranged to form the error correction model (ECM),

$$\Delta p_t = \alpha \beta_t' x_{t-1} + \sum_{i=1}^{m-1} \Gamma_i \Delta p_{t-i} + \varepsilon_t, \quad (12)$$

where Δ is the difference operator, $\beta_t' x_{t-1}$ is the cointegration vector, and x_{t-1} (p_{t-1}, d_{t-1}) comprises all endogenous and exogenous covariates as well as the constant term c of the price Equations (8) and (9). β_t' is a vector of the long-run cointegration parameters, whereas α is a parameter matrix that captures the speed at which deviations ($\text{ECT}_{t-1} = \beta_t' x_{t-1}$) from the equilibrium are corrected in each time period. If any element of α takes the value 1, deviations are immediately corrected. Although the error correction term (ECT) captures the long-run response, the Γ_i parameters measure the short-run response of Δp_t to shocks (Lloyd et al., 2009; Steen & Salvanes, 1999).

Although it is generally assumed that coefficients are time-invariant in ECMs, we allow the cointegration parameters (i.e. the β parameters from Equations (8) and (9)) to be time-variant, since the CEs are elements of these parameters. The following procedure is implemented. In an initial step, we apply Lütkepohl and Krätzig's (2004) simple two-step (S2S) estimation procedure. The α matrix is estimated separately from the other parameter matrices via ordinary least squares (OLS) by normalizing the cointegration vector. The normalization is accomplished by setting one of the cointegration coefficients, in this case the β of p^f , to the value of one. With the estimates of α , we can estimate a transformed equation in the second step, again via OLS, and obtain the cointegration parameters β (Lütkepohl & Krätzig, 2004). The Kalman-Filter is applied to estimate the transformed equation and retrieve estimates of the time-variant β coefficients. This procedure allows the estimated firms' conjecture to be dynamic and to vary at each point in time. The Kalman-Filter is a recursive procedure, in which the estimates of the unknown state, here the parameters, are updated in each time period with new observations on the observable data (Kalman, 1960).

For the estimation, we derive the following observation and state equation for our model,

$$\Delta p_t^* = \beta_t (\Theta_t)' x_{t-1}^* + \varepsilon_t^* \quad (\text{observation equation}), \quad (13)$$

$$\beta_t (\Theta_t)' = \beta_{t-1} (\Theta_t)' + \omega_t \quad (\text{state equation}), \quad (14)$$

where $*$ indicates the transformation of the corresponding ECM vectors.⁵ The error term ω_t is a vector of normal and identical distributed disturbances with zero mean and nondiagonal covariance matrix Ω . Applying the Kalman-Filter produces $k - 1$, the actual number depends on whether Equation (8) or (9) is estimated, β_t parameter estimates, because the cointegration parameter of p^f is set to one.

As mentioned in the previous section, the $k - 1$ time-variant cointegration parameters are specified as to being comprised of a time-invariant and a time-variant component. To separate the two components, dynamic factor analysis (DFA) is applied (Stock & Watson, 2005). The general idea of DFA is that a set of univariate time series forms a multivariate system of variables. Instead of trends unique to each individual univariate time series, the system responds to changes in common factors (Zuur, Fryer, Jolliffe, Dekker, & Beukema, 2003). With the assumption of one common dynamic factor $\lambda_t(\Theta_t)$ explaining the evolution over time of every β_t , the DFA takes a similar form as the state-space representation of the ECM (13) and (14),

$$\beta_t^r (\Theta_t) = \delta_c^r + \delta_\lambda^r \lambda_t (\Theta_t) + \xi_t \quad (\text{observation equation}), \quad (15)$$

$$\lambda_t (\Theta_t) = \lambda_{t-1} (\Theta_t) + \tau_t \quad (\text{state equation}), \quad (16)$$

where r is a subscript for the $k - 1$ estimated time-variant cointegration coefficients. The time-invariant coefficients δ_c^r and δ_λ^r are compounds of the parameters of the derived supply Equations (1) and (5), and demand Equations (4) and

⁵ $\Delta p_t^* = ((\alpha' \Sigma^{-1} \alpha) \alpha' \Sigma^{-1} (\Delta p_t - \alpha p_t^f)) \times (I - \Delta p_{t-1}' (\Delta p_{t-1} \Delta p_{t-1}')^{-1} \Delta p_{t-1})$;
 $x_{t-1}^* = (p_t^p, z_t^f, z_t^p, z_t^m, t^f, t^p, t^r) \times (I - \Delta p_{t-1}' (\Delta p_{t-1} \Delta p_{t-1}')^{-1} \Delta p_{t-1})$;
 $\varepsilon_t^* = (\alpha' \Sigma^{-1} \alpha) \alpha' \Sigma^{-1} \varepsilon_t$;

(7) (for details see Equations (A7) and (A9) in the Appendix), and in contrast to the common factor, vary between the β_t parameters. The errors of the observation Equation (15) and of the state Equation (16) are normally distributed.

The state-space representation is estimated with maximum likelihood (ML). Nevertheless, the log likelihood function cannot be optimized directly, because the dynamic factors are initially unknown. Instead the expectation-maximization (EM) algorithm is applied, an iterative procedure that “successively maximizes the conditional expectation of the complete data likelihood function” (Zuur et al., 2003: 668). Thereby, the ML estimates of so-called “hyperparameters” are obtained. These hyperparameters contain the variances of the error terms, the δ parameters, and the starting values of the dynamic factor and its variance. After determining the ML values of the hyperparameters the Kalman-Filter allows to estimate the dynamic factor and its variance (Shumway & Stoffer, 2000). The DFA is carried out separately for the estimated β_t parameters of the price Equations (8) and (9) by using the MARSS package in R (Holmes, Ward, & Wills, 2012). The DFA estimation results for the parameters λ_t and δ allow to uniquely identify Θ_t^p and Θ_t^r (see in the Appendix (A8) and (A10)).

4 | DATA

4.1 | German dairy value chain

The German dairy value chain is one of the most important sectors of the German agri-food industry. In 2011, German dairy farmers generated around € 10.3 billion of production value, which accounted for about 19.5% of the total domestic agricultural production value (€ 52.2 billion). The dairy industry itself produced total revenues of € 26.8 billion in the same year. Around 1/4 of its revenue was generated outside the domestic market. The overall share of milk and dairy products in overall German exports made up to 0.8% and in total agri-food exports 8.0%. The German dairy supply chain is also of great significance to the global milk and dairy product supply. In 2011, around 1/5 of all EU's raw milk (151.3 million t) was produced in Germany (30.3 million t), a production volume that made Germany the sixth largest global producer of milk after the United States and the BRIC countries (Brazil, Russia, India, and China). Furthermore, German dairies ranked 5th in milk for human consumption, 3rd in butter, and 2nd in cheese production worldwide and were the main supplier of cheese to the world market in 2011 (AMI, 2014).

The German dairy value chain can be split into three main stages. Dairy farmers produce raw milk, delivered almost entirely (96.7%) to domestic dairies. German dairies procure their inputs within a maximum radius of about 250 km and almost exclusively domestically (98.5%), because of the perishable nature of raw milk (BMLEV, 2013; BKartA, 2012). The dairies in turn sell their output to a variety of customers including retailers, wholesalers, food processors, and smaller costumers (such as restaurants, hospitals, or schools), or export overseas.

Since 2000, the German dairy value chain not only underwent restructuring processes, including concentration and consolidation, of different speed and scale, but also decisive policy changes, in particular the gradual lifting of the milk quota (Hirsch & Hartmann, 2014). In 2000, about 136,000 dairy farmers operated with approximately 4.5 million dairy cows (about 33 cows per farm). Although, the number of dairy farmers dropped rapidly by roughly 1/3 to 89,000 in 2011, the head count of dairy cows on German farms decreased only slightly, to 4.2 million. Thus, the average size of German dairy farms grew to around 47 dairy cows per farm (BMLEV, 2013). Nevertheless, these developments have been relatively heterogeneous; in 1999, approximately 79.2% of all dairy cows were held in farms with less than 100 heads, the share later decreased to 58.2% (BMLE, 2014).

Between the years 2000 and 2011, the number of German dairies declined by around 1/5 from originally 152 to 126 (BMLEV, 2013). The decline was mainly due to mergers and acquisitions among dairy companies in order to increase processing capacities, to implement cost-minimizing strategies, or to expand brand product portfolios (Theuvsen & Ebnet, 2005). In particular, mergers between large dairy cooperatives (e.g. Nordmilch and Humana Milch in 2011) have resulted in a highly concentrated dairy processing industry. In 2011, the six largest dairy processing companies processed 49.1% of German raw milk supply and accounted for 31.8% of the total dairy industry's revenues (BMLEV, 2013; 2014).

However, four of the six largest dairy companies (in terms of revenue) were dairy cooperatives. The primary aim of these integrations of farmer groups into the downstream market is to promote and strengthen farmers' bargaining power and thereby to receive higher raw milk prices for their members. Nevertheless, due to the recent EU policy changes promoting market liberalization, intrinsic organizational problems (e.g. the lack of internal capital that prevents innovations), and cost-minimizing strategies, dairy cooperatives often are not able to obtain a high raw milk price. Instead, the cooperatives have been trying to procure raw milk at the lowest price possible (BKartA, 2009; Grau, Hockmann, & Levkovych, 2015). Consequently, the large differences in the numbers of suppliers and buyers together with the procurement strategies of dairy cooperatives have raised dairy farmers' concerns regarding the existence of market power in the German raw milk market.

At a first glance, the downstream market for dairy products appears not to be characterized by power imbalances. Buyers of dairy products seem numerous and heterogeneous, comprised of few large food retailers, processors, and thousands of restaurants and food-service establishments (BKartA, 2009). However, the majority of dairy products are actually traded by less than a handful of retailers. In the case of milk for human consumption, only three retailers, Aldi, Rewe, and Edeka, cover 50–65% of the entire demand. Furthermore, the main buyer of German dairy exports is essentially the German retail sector; as the domestic dairy industry serves as the primary supply base for European subsidiaries of German retailers (Friedrich, 2010; Hellberg-Bahr et al., 2010). In recent years, procurement cooperations have been established between the dominant and smaller retailing chains with market shares of less than 5%, a development further boosting the overall procurement share of individual retailers. Consequently, a few retailers control almost the entire domestic and international market of German dairy products. All these factors increase the threat of oligopsonistic market conduct by retailers on the German dairy output market (BKartA, 2012).

Given the concentration and consolidation processes at the processing and retail level, it is not surprising that, since the early 2000s, the German Federal Cartel Office has received a growing number of complaints by farmers and dairies regarding the procurement behavior of downstream buyers (dairy industry and retail sector). Dairy farmers have been complaining about dairies' asymmetric price transmission (i.e., price decreases are transmitted faster than price increases). German dairies have been accusing retailers of abusing their bargaining position to lower the prices for dairy products. As a result, the German FCO started an official investigation in 2008. In the investigation's final report, published in 2012, the German FCO established that, although procurement prices were low and the structure along the chain seemed to favor the position of buyers, there was no evidence of the abuse of market power (BKartA, 2009, 2012).

4.2 | Descriptive data statistics and cointegration analysis

The time period of our study spans from January 2000 to March 2011. Table 1 presents descriptive statistics on all price time series used for estimation. All time series were deflated using the national Consumer Price Index (CPI). Although data on the raw milk price is publicly available in form of the average milk price at farm-gate, the dairy product price had to be constructed as a unit-share weighted average price of major dairy products. It consists of the wholesale prices for Emmentaler, SMP, and German brand butter (block form).⁶ In addition, the database includes all available relevant prices for the nonspecialized inputs for the production and marketing process along the dairy value chain. The largest cost factor at the farm level is feed. The procurement of feed summed up to around 46.9% of the intermediate consumption in the overall German agricultural production in 2014 (BMLE, 2016). At the processing level, material input, i.e. raw milk, and labor costs (average wage in the German food industry), accounted for the largest share in total cost with respectively 68.8% and 6.7% (BMLEV, 2013). Aside from the procurement price for dairy products, the principal cost item at the retail level, in general, are wages (Lloyd et al., 2009), e.g. accounting for up to 30% of the US retail prices for dairy products (Hovhannisyan & Gould, 2012). Therefore, the average wage of German retail employees was included as the nonspecialized input at the retail level. Relevant data were available at a monthly frequency for all but two time series. Data of the wages for employees in the food industry and in retailing were only available with a

⁶ The conversion rates are 25, 10, and 7.5 L/kg, respectively.

TABLE 1 Descriptive statistics of data (January 2000 to March 2011) in €

Item	Series	Symbol	Min.	Max.	Mean	Source	Notes
Milk farm gate price	Monthly data	p^f	0.20	0.38	0.30	AMI/ZMP	Seasonally adjusted using x12arima
Implied processors' milk price	Monthly data for SMP, Butter, and Emmentaler	p^p	0.33	0.56	0.43	AMI/ZMP	Calculated using technical conversion factors and shares on processing
Feed price	Concentrate for milk production, monthly data	z^f	0.14	0.23	0.17	AMI/ZMP	
Average wage food industry	Quarterly data on Salaries and labor input	z^p	29.59	32.63	30.76	Statistisches Bundesamt	Interpolation to monthly data
Average wage retail employee	Quarterly data on salaries and labor input	z^r	19.57	20.98	20.31	Statistisches Bundesamt	Interpolation to monthly data

Source: Own elaboration.

TABLE 2 Unit root (KPSS Test), cointegration tests, and Chow forecast test

KPSS Test	Test Statistics
Milk farm gate price p^f	2.6471 ^{***a}
Implied processors' milk price p^p	4.0518 ^{***a}
Johansen Trace Test	
Lag length	3 ^b /4 ^c
Rank 0 ^d	38.83 ^{***} /48.46 ^{***}
Rank 1 ^e	15.38 ^{**} /16.86 ^{***}
Chow forecast test (with 3 lags)	
Equation (8)	0.00/0.04
Equation (9)	0.00/0.05

Source: own elaboration.

Notes:

^{**}Significance at the 5% level.

^{***}Significance at the 1% level.

^aCritical value to reject hypothesis of unit root at 5% level 0.463.

^blag length selection according to the Schwarz Criterion.

^clag length selection according to the Akaike Info Criterion.

^dCritical values at 5% level 25.73/12.45.

^eCritical values at 5% level 15.76/6.79.

^fAfter 100 bootstrap replications and search over all data points.

quarterly frequency. For consistency, these two price series were interpolated to monthly frequencies. The inclusion of capital costs and energy costs would have been a helpful addition to the analysis; however, data on these cost factors were only available on a yearly basis, and thus these variables not included in the analysis.⁷

Table 2 illustrates the results of the tests for stationarity, cointegration, and the Chow forecast test. The results of the unit root test via the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test with level stationarity (Kwiatkowski, Phillips, Schmidt, & Shin, 1992) show that not only for the time series of the seasonal adjusted farm price, but also for the dairy output price the hypothesis of stationarity has to be rejected. Thus, the presence of a unit root is not rejected.

⁷ This decision probably did not significantly affect the analyses, since these costs, particularly at the dairy level, are negligible in total costs. Energy consumption only made up 1.9% and capital consumption 1.7% of the overall production costs of dairies (BMLEV, 2013).

The next step was to test for cointegration among the endogenous variables, the price for raw milk and the price for dairy output. However, the other cost variables are also elements of the price equations. The elements enter the cointegration relation as deterministic terms, but are exogenous to the VECM, similar to the trend and the constant terms (see Equations (11) and (12)). These variables do not react to past developments of the endogenous variables, or to shocks. The hypotheses of rank 0 and rank 1, i.e. one cointegration relationship, were not statistically rejected by the Johansen trace test (Johansen, 1991). Hence, we continued in our empirical application with the assumption of one cointegration process.

The stability of the long-run cointegration parameters was tested via the Chow forecast test, under the null hypothesis of time-invariant parameters and residual covariance matrix against the alternative that these may vary (Lütkepohl & Krätzig, 2004). The results of the test confirm our assumption that the parameters are not time-invariant, because the null hypothesis is rejected at the 5% significance level or higher, at all time periods for both equations (see Table 2).

As suggested by the results of the tests, the estimations of Equations (13) and (14) were carried out separately under the assumptions of one cointegration relationship and time-variant long-run cointegration parameters. Before estimating the transformed VECMs of both price equations, (8) and (9), the optimal lag length of the differenced variables to be included had to be determined. Therefore, the Akaike Information Criterion (AIC) and the Schwarz Criterion (SC) were applied. While the result of the AIC suggests including a lag length of three for Equation (8) and four for Equation (9), the SC only proposes three lags for both price equations (see Table 2; Lütkepohl & Krätzig, 2004). Because the criteria differ with respect to the lag length, and in order not to miss the model specification that best fits the data, VECMs with a lag length of one to four were specified and estimated for both price equations. The next section presents the model selection procedure as well as the results for the time-variant cointegration parameters, the DFA, and the CEs of the best fitting specifications.

5 | RESULTS AND DISCUSSION

Table 3 displays the ΔAICc and the weighted AICc of the DFA, on the basis of which the model with the lag length that best fits the data can be chosen. The results clearly indicate that for price Equation (8), the specification with one differenced lag of the endogenous price variable is the one fitting best the data, whereas for the second price Equation (9) the best fitting specification includes a lag length of two. Consequently, the empirical results presented in the following sections are based on the following VECM setups: one cointegration vector among the endogenous price variables, time-variant cointegration parameters, and a lag length of one and two for the Equations (8) and (9), respectively.

Table 4 reports the estimation results for the time-variant cointegration coefficients of the price Equations (8) and (9). Because all coefficients show variations and are statistically significant at least at the 5% significance level over the entire investigated period, all time-variant parameter estimates enter the DFA for the price Equations (8) and (9) as dependent variables.

The cointegration coefficient estimates were then separated into a time-invariant and a dynamic component, λ_t^p and λ_t^r , respectively, using DFA (see (A7) and (A9) in the Appendix). The DFA parameter estimates and the corresponding

TABLE 3 List of model specifications differing in lag length and their DFA results for AICc and the weighted AICc

No. of differenced lags	Equation (8)		Equation (9)	
	ΔAICc	Weighted AICc	ΔAICc	Weighted AICc
1	0	100%	197.957	0%
2	369.727	0%	0	100%
3	597.406	0%	231.994	0%
4	795.424	0%	259.611	0%

Source: Own elaboration, AICc values were obtained from the estimations within the MARSS package in R.

TABLE 4 Results and average standard errors of state space models estimations (β -parameters)

Parameter	Equation (8)				Equation (9)			
	Min.	Max.	Mean	Ø S.E.	Min.	Max.	Mean	Ø S.E.
p^p	-0.9226	-0.8411	-0.8775***	0.1087	-0.6619	-0.5933	-0.6189***	0.0702
z^f	0.4882	0.6978	0.5812***	0.1656				
z^p	0.0245	0.0298	0.0275***	0.0071				
z^r					-0.0285	-0.0166	-0.0218**	0.0098
Constant	-0.8608	-0.6904	-0.7984***	0.2393	-0.5996	-0.3309	-0.4814**	0.2238
Trend	-0.0015	-0.0006	-0.0010***	0.0003	-0.0008	-0.0006	-0.0007***	0.0001

Source: Own elaboration.

Notes:

**Significance at the 5% level.

***Significance at the 1% level.

TABLE 5 DFA estimation results and standard errors for the constant parameters

Parameter	Value	Standard Error	Parameter	Value	Standard Error
ω_c^{pp}	-0.7708***	0.0542	δ_c^{pp}	-0.6064***	0.0275
ω_c^{zf}	0.2358	0.1555	δ_c^{zf}	-0.0479***	0.0018
ω_c^{zp}	0.0180	0.0673	δ_c^t	-0.0009***	0.0234
ω_c^t	0.0002	0.0647	δ_c^c	0.0946**	0.0412
ω_c^c	-0.5004***	0.0301	δ_λ^{zf}	-2.0878***	0.2421
ω_λ^{pp}	0.3351**	0.1430	δ_λ^t	-0.0177***	0.0023
ω_λ^{zf}	-1.0852**	0.5212	δ_λ^c	46.1341***	5.4335
ω_λ^{zp}	-0.0300	0.2155			
ω_λ^t	0.0039	0.2060			
ω_λ^c	0.9362***	0.0984			

Source: Own elaboration.

Notes:

**Significance at the 5% level.

***Significance at the 1% level.

TABLE 6 DFA estimation results and average standard errors for the dynamic factors

Parameter	Min.	Max.	Mean	Ø S.E.
λ_t^p	-0.4042	-0.2515	-0.3183***	0.0058
λ_t^r	-0.0148	-0.0095	-0.0125***	0.0001

Source: own elaboration.

Note:

***Significance at the 1% level.

standard errors for the two estimated equations are reported in Tables 5 and 6. The dynamic factors, depending on the CEs of the processors and retailers, respectively, vary considerably over the data period, and are statistically significant at the 1% level for all time periods. The estimated time-invariant parameters show a mixed picture regarding their significance (see Table 5). However, all parameters necessary for the calculation of the CEs are statistically significant at the 1% level. With the results of the DFA, we can calculate the CEs of the processing industry as well as that of retailers (see in the Appendix (A8) and (A10) for details on how this calculation is performed).

The calculated CEs are consistent with theory and lie between 0 and 1 (see Fig. 1). Large variations in their values are observable in the data period. Although processors' CE varies between 0.04 and 0.07, that of retailers' ranges from 0.29 to 0.39.

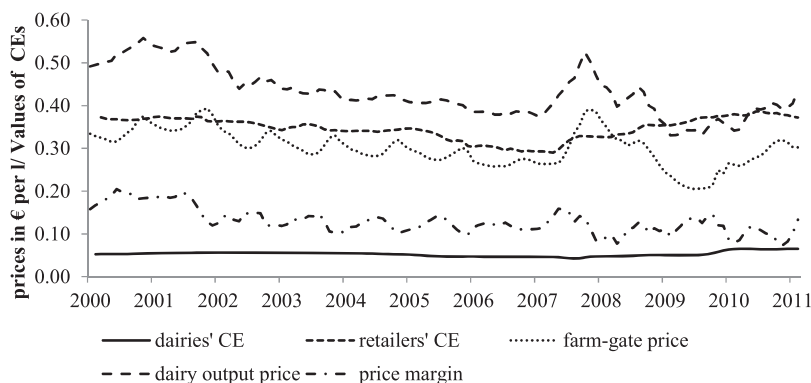


FIGURE 1 Values of the conjectural elasticities of the dairy industry Θ_t^p and retailers Θ_t^r , as well as German milk farm gate price, share weighted dairy output price and the price margin from January 2000 to March 2011. Source: Own elaboration

The estimated CEs indicate some degree of oligopsonistic market conduct by downstream actors in both markets. Nevertheless, the CEs are in both cases far from monopsony values.

The raw milk market, with values between 0.04 and 0.07, is close to perfectly competition. The rather low values of oligopsonistic market conduct of the raw milk market are comparable to other studies, which have investigated the raw milk markets in other European countries. Hockmann and Vőneki (2009) also found a slightly imperfect market (CE of around 0.1) for raw milk in Hungary over the period 1999–2006. Perekhozhuk, Hockmann, Fertő, and Bakucs (2013) who also examined the Hungarian raw milk market, but using firm level data, reported higher values of oligopsonistic CE (between 0.22 and 0.30) from 1993 to 2006. Perekhozhuk, Glauben, Teuber, and Grings (2015) analyzed firm-level data of the Ukrainian raw milk market and estimated a national CE value of 0.15 and values between 0.09 and 0.32 for regional markets for the period from 1996 to 2015. In contrast, Zavelberg, Wieck, and Heckeleei (2015) applied a regional approach to the German raw milk market from 2001 to 2012 and found indications for oligopsonies in both the national market and the regional ones, with CEs around 0.75. The differences in estimates might be explained by the fact that regional markets (large dairies, such as the Deutsche Milch Kontor [DMK] effectively managing more than 50% of regional supply [BKartA, 2009]) frequently show a more concentrated buyers' structure than national ones. Also, different from the model presented in this paper, Zavelberg et al. (2015) assumed that retailers do not exercise market power.

Although the European raw milk markets have been generally characterized as competitive, the low level of oligopsonistic conduct found in the German raw milk is surprising, mainly because of the high concentration of the processing industry. However, dairies, in particular cooperatives, often adopt a cost-minimizing strategy and aim to fully utilize their production capacity at all times (Richards, Patterson, & Acharya, 2001). This strategy can only be accomplished by having a constant, guaranteed supply of raw milk. Consequently, the exertion of market power might provide higher profits in the short-run, but, in the long-run, might have detrimental effects, such as a permanent decline in supply quantity due to low prices and lower incentives for farmers to invest, which will lead to lower overall profits (Crespi, Saitone, & Sexton, 2012). Furthermore, cooperatives really seem able to break the bargaining position of private dairies and thus foster competition (Salhofer et al., 2012).

Larger levels of market power were observed in the German dairy output market. The estimated CE of retailers ranges from 0.29 to 0.39, values that are far from monopsony levels, but are larger than those estimated for processors. The previously mentioned studies on retailers' oligopsony power unfortunately do not provide any comparable results, since they either only reported a Lerner Index (Gohin & Guyomard, 2000) or analyzed non-dairy product markets (Anders, 2008). Salhofer et al. (2012) is the only study that provided a suitable basis for comparison. The extent of retailers' oligopsony power in the Austrian butter market was estimated to be significantly lower at around 0.10. The difference in the degree of competition between the Austrian and German dairy product market might be explained

by the intensive horizontal competition of the five large German retailers in consumer markets. Because the consumer markets are highly competitive, the exertion of oligopsony power might be the only way by retailers to increase profits or market shares by offering lower prices, initially secured by significantly marking-down procurement prices (Anders, 2008).

The evolution of the CEs for both processors and retailers are similar: after a rather stable period between 2000 and 2006, the CEs of dairies and retailers dropped during the period of high prices (2007/2008, see Fig. 1), due to empty public EU milk powder stocks as well as sharp increases in feed prices due to low global supply and high demand for grains from the biofuel industry (Acosta, Ihle, & Robles, 2014). During this period German dairies and retailers, but in particular the latter, may have acted less collusively in order to expand procurement in a period of stable supply (Perloff, Karp, & Golan, 2007). After the price peak period, the CE of retailers recovered more quickly than that of dairies. It appears that retailers returned to using their market power to lower procurement prices when supply was starting to grow due to the gradual lifting of the milk quota. The dairies, although with rather low levels of market power, either restrained from this action for longer or were limited in their ability to exert market power due to their processing obligations towards dairy farmers.

Interestingly, the price margin between dairy output price and raw milk price also spiked around 2007, after years of lower differences (see Fig. 1). It seems that dairies did not transmit immediately the price increases on the downstream market to dairy farmers, allowing dairy processors to earn extra profits for a brief period of time. In contrast, during a period of continuous decline in dairy output price, from 2002 to 2006, the margin between dairy output price and raw milk was rather stable. However, the relatively high difference between the prices could not be sustained for a long period. It dropped to an all-time low in 2008, and afterwards leveled out around a lower level until the end of the investigated period.

When Vóneki, Mándy-Nagy, and Stark (2015) estimated the raw milk supply of farmer after the final abolishment of the quota, they found the supply growing further. Given that many cooperatively organized dairies are obliged to process any milk delivered by their members, we expect the supply of dairy output to expand as well. The excessive supply of dairy outputs will promote the bargaining position of the retail sector. Additionally, retailers have developed marketing strategies, such as own private labels, to gain further control over price transmissions (Loy, Weiss, & Glauben, 2016). In the raw milk market, a growing supply might seem to strengthen the bargaining position of dairies at first glance. However, since dairy processors are obliged to accept all delivered milk, dairies not only lose their flexibility in raw milk procurement but also the ability to use material input supply as a strategic variable. Thus, despite minimal increases at the end of the investigation period, the CE of processors is likely to remain low, or vanish completely in the course of the liberalized milk market.

6 | CONCLUSIONS

In this paper, we present a novel approach to estimate time-variant oligopsony power in the successive markets of an agri-food value chain. A NEIO structural model of the input and output markets of a value chain laid the foundation for deriving two price equations that allow the independent identification of oligopsony power in the upstream and downstream market of a value chain without quantity data. The approach was applied to the German dairy value chain, and provides new evidence on market power in the German raw milk and dairy product markets. To the best of our knowledge, there are no attempts to model oligopsony power in two successive vertically integrated markets. The markets are modeled in the fashion of NEIO, but the derivations of partial equilibrium equations and the final pricing equations significantly lower the data requirement and the number of equations to be estimated. Moreover, by assuming time-variant oligopsony conduct and applying the novel estimation strategy, the estimated market power indices are modeled to adjust in each time period, thus being dynamic rather than being "static."

The estimation procedure combined three time series estimation techniques. The vector error correction model (VECM) is transformed via the simple two step estimation technique of Lütkepohl and Krätzig (2004). Then, the Kalman-Filter is applied to the transformed VECM to obtain the time-variant parameters. The time variation is

assumed to be caused by changes in oligopsony conduct and is extracted in form of dynamic factors through the application of DFA to the VECM's parameter. In addition, the DFA also allows separating the dynamic factors from other time-invariant components. The results of the DFA permit to uniquely identify the time-variant market power indices.

The model is applied to the German dairy value chain over the period January 2000–March 2011. The results show the existence of oligopsonistic behavior in both the raw milk market and the dairy output market, between monopsony and perfect competition, albeit close to perfect competition in the case of the raw milk market. The development over time of the conjectural elasticities of processors and retailers are similar, being relatively stable in the 2000–mid-2005 period and then reaching a minimum in mid-2007 after a period of rapid decline. Retailers' oligopsony power recovered faster than that of dairies' and, in 2011, regained the same level as in 2000, at the beginning of the investigation period. In contrast, the dairies' market power is larger at the end of the data period than it was at the beginning. The lower levels of oligopsonistic behavior during the period from 2006 to 2009 are probably a result of the global food price and demand peaks.

Although we were able to estimate time-variant oligopsonistic behavior along the German dairy value chain, our approach has limitations. First, through the assumption of a quantity setting game, we are able to estimate the average input conjectural elasticity and thus the level of oligopsonistic behavior, but we cannot make any inference about the actual game played by the firms (Sckokai, Soregaroli, & Moro, 2013). Second, the assumption that exports and imports are negligible might be realistic for the raw milk market, but not for processed dairy products, as the German dairy industry has been expanding on global markets. However, we follow other NEIO studies in which the imports and exports are frequently excluded from the analysis due to lack of data (Sckokai et al., 2013). Finally yet importantly, the homogenous product assumption is very restrictive. Although raw milk seems to be a quite homogenous product market, the aggregation across dairy products is at least questionable. Because of the lack of available data, the unit-share weighted dairy output price was calculated using prices determined in the German dairy stock market for generic products. Nevertheless, the German dairy product market also includes a vast majority of heterogenous brand products, which achieve tremendously high prices. Although the obtained results might well describe the extent of market power for the generic dairy products, the extent of market power in brand dairy product markets might be lower (Perloff et al., 2007).

Although the assumptions made in this paper might limit the validity of the obtained explanatory power of the obtained results, they were necessary to reduce the complexity of the theoretical model and because of data limitations. However, in future research it would be desirable to relax some of these assumptions. Instead of assuming that firms optimize their profits in each single time period, future research could allow firms to maximize a discounted stream of profits and consider also long-run strategic variables, e.g. firms' investments in processing capacities (Perloff et al., 2007). Furthermore, future research could model successive oligopsonies including also exports and imports, or relax the restrictive assumption of modeling the dairy chain as a homogenous product market. Similar to the works of Villas-Boas and Hellerstein (2006) and Cakir and Balagtas (2012), our approach could be extended to incorporate a homogenous product upstream market followed by a downstream market with differentiated products.

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REFERENCES

- Acosta, A., Ihle, R., & Robles, M. (2014). Spatial price transmission of soaring milk prices from global to domestic markets. *Agribusiness*, 30(1), 64–73.
- Agrarmarkt Informations-Gesellschaft (AMI) (different volumes). *AMI Markt Bilanz Milch*. Bonn, Germany: AMI GmbH.
- Anders, S. M. (2008). Imperfect competition in German food retailing. *Atlantic Economic Journal*, 36(4), 441–454.
- Bell, R., Davies, R., & Howard, E. (1997). The changing structure of food retailing in Europe: the implications for strategy. *Long Range Planning*, 30(6), 853–861.

- Bresnahan, T. F. (1989). Empirical studies of industries with market power. In R., Schmalensee & R. D., Willig (Eds.), *Handbook of industrial economics 2* (pp. 1011–1057). Amsterdam, The Netherlands: Elsevier B.V.
- Bundeskartellamt (BKartA) (2012). *Sektoruntersuchung Milch - Abschlussbericht gemäß § 32e GWB - Januar 2012*. Bonn, Germany: Bundeskartellamt.
- Bundeskartellamt (BKartA) (2009). *Sektoruntersuchung Milch. Zwischenbericht*. Bonn, Germany: Bundeskartellamt.
- Bundesministerium für Ernährung, Landwirtschaft und Verbraucherschutz (BMLEV) (different volumes). *Statistisches Jahrbuch über Ernährung, Landwirtschaft und Forsten der Bundesrepublik Deutschland*. Münster-Hiltrup, Germany: Landwirtschaftsverlag GmbH.
- Bundesministerium für Ernährung und Landwirtschaft (BMLE) (different volumes). *Statistisches Jahrbuch über Ernährung, Landwirtschaft und Forsten der Bundesrepublik Deutschland*. Münster-Hiltrup, Germany: Landwirtschaftsverlag GmbH.
- Cakir, M., & Balagtas, J. V. (2012). Estimating market power of U.S. Dairy Cooperatives in the fluid milk market. *American Journal of Agricultural Economics*, 94(3), 647–658.
- Canning, P. (2011). *A revised and expanded food dollar series: A better understanding of our food costs*. ERR-114, U.S. Department of Agriculture, Economic Research Service.
- Chambers, R. G. (1988). *Applied production analysis: A dual approach*. Cambridge, UK: Cambridge University Press.
- Clarke, R., Davies, S. W., Dobson, P. W., & Waterson, M. (2003). Buyer power and competition in European food retailing. *Review of Industrial Organization*, 22(4), 337–340.
- Consultative Commission on Industrial Change (CCMI) (2008). The evolution of the high-volume retail sector in Europe over the past 5 years—final report. Available at Retrieved from https://www.eesc.europa.eu/?i=portal.en.industrial-sectors-retail-external-studies.6534with_annex.doc (accessed 24 May 2016).
- Corts, K. S. (1999). Conduct parameters and the measurement of market power. *Journal of Econometrics*, 88(2), 227–250.
- Crespi, J. M., Saitone, T. L., & Sexton, R. J. (2012). Competition in the U.S. farm product markets: Do long-run incentives trump short-run market power? *Applied Economic Perspectives and Policy*, 34(4), 669–695.
- Dobson, P. W., Waterson, M., & Davies, S. W. (2003). The patterns and implications of increasing concentration in European food retailing. *Journal of Agricultural Economics*, 54(1), 111–125.
- Dockner, E. J. (1992). A dynamic theory of conjectural variations. *The Journal of Industrial Economics*, 40(4), 377–395.
- European Commission (EC) (2009). *Analysis of price transmission along the food supply chain in the EU. SEC(2009) 1450*. Brussels, Belgium: Commission of the European Communities.
- Fałkowski, J. (2010). Price transmission and market power in a transition context: Evidence from the Polish fluid milk sector. *Post-Communist Economics*, 22(4), 513–529.
- Friedman, J. W. (1983). *Oligopoly theory*. Cambridge, UK: Cambridge University Press.
- Friedrich, C. (2010). *Milchverarbeitung und -vermarktung in Deutschland – eine deskriptive Analyse der Wertschöpfungskette. Arbeitsberichte aus der vTI- Agrarökonomie*. Braunschweig, Germany: Johann Heinrich von Thünen-Institut.
- Genesove, D., & Mullin, W. P. (1998). Testing oligopoly models: Conduct and cost in the sugar industry, 1890–1914. *The RAND Journal of Economics*, 29(2), 355–377.
- Gohin, A., & Guyomard, H. (2000). Measuring market power for food retail activities: French evidence. *Journal of Agricultural Economics*, 51(2), 181–195.
- Gollop, F. M., & Roberts, M. J. (1979). Firm interdependence in oligopolistic markets. *Journal of Econometrics*, 10, 313–331.
- Grau, A., Hockmann, H., & Levkovych, I. (2015). Dairy cooperatives at the crossroads. *British Food Journal*, 117(10), 2515–2531.
- Hellberg-Bahr, A., Pfeuffer, M., Steffen, N., Spiller, A., & Brümmer, B. (2010). *Preisbildungssysteme in der Milchwirtschaft – Ein Überblick über die Supply Chain Milch*. Göttingen, Germany: Department für Agrarökonomie und Rurale Entwicklung, Universität Göttingen.
- Hendry, D. F., & Doornik, J. A. (2001). *Modelling dynamic systems using PcGive 10*. Vol. 10 II. London, UK: Timberlake.
- Hirsch, S., & Hartmann, M. (2014). Persistence of firm-level profitability in the European dairy processing industry. *Agricultural Economics*, 45(S1), 53–63.
- Hockmann, H., & Vónéki, É. (2009). Collusion in the Hungarian market for raw milk. *Outlook on Agriculture*, 38(1), 39–45.
- Holmes, E. E., Ward, E. J., & Wills, K. (2012). MARSS: Multivariate autoregressive state-space models for analyzing time-series data. *The R Journal*, 4(1), 11–19.
- Hovhannisyan, V., & Gould, B. W. (2012). A structural model of the analysis of retail market power: The case of fluid milk. *American Journal of Agricultural Economics*, 94(1), 67–79.

- Johansen, S. (1991). Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models. *Econometrica*, 59(6), 1551–1580.
- Juselius, K. (2006). *The cointegrated VAR Model*. Oxford, UK: Oxford University Press.
- Kalman, R. E. (1960). A new approach to linear filtering and prediction problems. *Journal of Basic Engineering, Transactions ASMA, Series D82*, 35–45.
- Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., & Shin, Y. (1992). Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics*, 54(1–3), 159–178.
- Lloyd, T., McCorrison, S., Morgan, W., Rayner, A., & Weldegebril, H. (2009). Buyer power in the U.K. Food Retailing: A 'Frist-Pass' test. *Journal of Agricultural & Food Industrial Organization*, 7(1). Article 5.
- Loy, J.-P., Weiss, C. R., & Glauben, T. (2016). Asymmetric cost pass-through? Empirical evidence on the role of market power, search and menu costs. *Journal of Economic Behavior & Organization*, 123, 184–192.
- Lütkepohl, H., & Krätzig, M. (2004). *Applied time series econometrics*. Cambridge, UK: Cambridge University press.
- Madau, F. A., Furesi, R., & Pulina, P. (2016). The existence of buyer power in the Italian fresh milk supply chain. *British Food Journal*, 118(1), 70–82.
- Muth, M. K., & Wohlgenant, M. K. (1999). Measuring the degree of oligopsony power in the beef packing industry in the absence of marketing input quantity data. *Journal of Agricultural and Resource Economics*, 24(2), 299–312.
- Nevo, A. (2001). Measuring market power in the ready-to-eat cereal industry. *Econometrica*, 69, 307–342.
- Nevo, A. (1998). Identification of the oligopoly solution concept in a differentiated-products industry. *Economics Letters*, 59, 391–395.
- Perekhozhuk, O., Glauben, T., Teuber, R., & Grings, M. (2015). Regional-level analysis of oligopsony power in the Ukrainian Dairy Industry. *Canadian Journal of Agricultural Economics*, 61(1), 43–76.
- Perekhozhuk, O., Hockmann, H., Fertő, I., & Bakucs, L. Z. (2013). Identification of market power in the Hungarian Dairy Industry: a plant-level analysis. *Journal of Agricultural & Food Industrial Organization*, 11(1), 1–13.
- Perloff, J. M., Karp, L. S., & Golan, A. (2007). *Estimating market power and strategies*. Cambridge, UK: Cambridge University Press.
- Richards, T. J., Patterson, P. M., & Acharya, R. N. (2001). Price behavior in a dynamic oligopsony: Washington processing potatoes. *American Journal of Agricultural Economics*, 83(2), 259–271.
- Rojas, C. (2008). Price competition in U.S. brewing. *The Journal of Industrial Economics*, 56(1), 1–31.
- Salhofer, K., Tribl, C., & Sinabell, F. (2012). Market power in Austrian food retailing: The case of milk products. *Empirica*, 39(1), 109–122.
- Sckokai, P., Soregaroli, C., & Moro, D. (2013). Estimating market power by retailers in a dynamic framework: the Italian PDO cheese market. *Journal of Agricultural Economics*, 64(1), 33–53.
- Schroeter, J. R. (1988). Estimating the degree of market power in the beef packing industry. *The Review of Economics and Statistics*, 70(1), 158–162.
- Sexton, R. J. (2013). Market power, misconcepts, and modern agricultural markets. *American Journal of Agricultural Economics*, 95(2), 209–219.
- Sexton, R. J. (2000). Industrialization and consolidation in the U.S. Food Sector: Implications for competition and welfare. *American Journal of Agricultural Economics*, 82(5), 1087–1104.
- Shumway, R. H., & Stoffer, D. S. (2000). *Time series analysis and its applications*. New York, USA: Springer.
- Slade, E. (2004). Market power and joint dominance in U.K. Brewing. *The Journal of Industrial Economics*, 52(1), 133–163.
- Sperling, R. (2002). Estimating oligopsony power in the United States market for slaughter hogs: An error correction approach. Electronic Thesis or Dissertation available at: Retrieved from <https://etd.ohiolink.edu/> (accessed 30 December 2016).
- Statistisches Bundesamt (2016). Bruttoverdienste, Wochenarbeitszeit: Deutschland, Quartale, Wirtschaftszweige, Leistungsgruppen, Geschlecht. Available at: Retrieved from https://www-genesis.destatis.de/genesis/online/data.jsessionid=7DA5BD39D4719_C518A8F99A584C256D7.tomcat_GO_1_3?operation=abruftabelleBearbeiten&levelin-dex=2&levelid=1482934850653&auswahloperation=abruftabelleAuspraegungAuswaehlen&auswahlverzeichnis=ordnungsstruktur&auswahlziel=werteabruf&selectionname=62321-0001&auswahltext=&werteabruf=starten (accessed 28 December 2016).
- Steen, F., & Salvanes, K. G. (1999). Testing for market power using a dynamic oligopoly model. *International Journal of Industrial Organization*, 17, 147–177.
- Stock, J. H., & Watson, M. W. (2005). Implications of dynamic factor models for VAR analysis. NBER Working Papers 11467.

- Theuvsen, L., & Ebneht, O. (2005). Internationalization of cooperatives in the agribusiness: Concepts of measurement and their application. In T. Theurl & E. C. Meyer (Eds.), *Strategies for cooperation* (pp. 395–419). Aachen, Germany: Shaker.
- Villas-Boas, S., & Hellerstein, R. (2006). Identification of supply models of retailer and manufacturer oligopoly pricing. *Economics Letters*, 90(1), 132–140.
- Vőneki, É., Mándy-Nagy, D., & Stark, A. (2015). Prospects for the European Union and Hungarian dairy sectors after the abolition of the milk quota system. *Studies in Agricultural Economics*, 117, 1–9.
- Zavelberg, Y., Wieck, C., & Heckelet, T. (2015). *How can differences in German raw milk prices be explained? 2015 Agricultural & Applied Economics Association and Western Agricultural Economics Association Annual Meeting*, San Francisco, CA, July 26–28.
- Zentrale Markt-und Preisberichtsstelle (ZMP) (different volumes). *ZMP-Marktbilanz Milch*. Bonn, Germany: Zentrale Markt-und Preisberichtsstelle GmbH.
- Zuur, A. F., Fryer, R. J., Jolliffe, I. T., Dekker, R., & Beukema, J. J. (2003). Estimating common trends in multivariate time series using dynamic factor analysis. *Environmetrics*, 14, 665–685.

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APPENDIX

(A1) Derived inverse supply function of raw milk:

$$p^f = \frac{\partial C(x^f, z_h^f, t^f)}{\partial x^f} = \rho^x + \sum_{i=1}^h \rho_i^{zx} z_i^f + \rho^{tx} t^f + \rho^{xx} x^f.$$

(A2) Aggregated revenue function of the processing industry:

$$\begin{aligned} R^p = & \varphi^0 + \varphi^p p^p + \varphi^{pp} \frac{p^{p2}}{2} + \varphi^x x^f + \varphi^{xx} \frac{x^{f2}}{2} + \sum_{j=1}^k \varphi_j^{zz} z_j^p + \sum_{j=1}^k \varphi_j^{zz} \frac{z_j^{p2}}{2} + \varphi^t t^p + \varphi^{tt} \frac{t^{p2}}{2} + \varphi^{px} p^p x^f + \sum_{j=1}^k \varphi_j^{pz} z_j^p p^p + \varphi^{pt} p^p t^p \\ & + \sum_{j=1}^k \varphi_j^{xz} z_j^p x^f + \varphi^{xt} x^f t^p + \sum_{\substack{i=1 \\ i \neq j}}^k \sum_{j=1}^k \varphi_{ij}^{zz} z_i^p z_j^p + \sum_{j=1}^k \varphi_j^{zt} z_j^p t^p. \end{aligned}$$

(A3) Derived demand function for raw milk:

$$p^f + \rho^{xx} x^f \Theta_t^p = \varphi^x + \varphi^{px} p^p + \varphi^{xx} x^f + \sum_{j=1}^k \varphi_j^{xz} z_j^p + \varphi^{xt} t^p.$$

(A4) Derived inverse supply function of dairy output:

$$p^p = - \frac{-y^p + \varphi^p + \varphi^{px} x^f + \sum_{j=1}^k \varphi_j^{pz} z_j^p + \varphi^{pt} t^p}{\varphi^{pp}}.$$

(A5) Derived demand function for dairy output:

$$p^p + \frac{y^p \Theta_t^r}{\varphi^{pp}} = g^y + g^{yy} y^p + \sum_{k=1}^m g_k^{zy} z_k^f + g^{ty} t^r.$$

(A6) Explicit solutions for the endogenous variables x^f , p^f , y^p , and p^p :

$$x^f = -\frac{1}{\varphi^{px}} \left(B - \frac{\varphi^{pp} E}{\Theta_t^r - \varphi^{pp} g^{yy}} + \frac{(-1 - \Theta_t^r + \varphi^{pp} g^{yy}) \left((A + BZ) \left(-\frac{\Theta_t^r}{\varphi^{pp}} + g^{yy} \right) + EZ \right)}{(\Theta_t^r - \varphi^{pp} g^{yy}) D} \right),$$

$$p^f = \sum_{i=1}^h \rho_i^{zx} z_i^f + \rho^{tx} t^f + \sum_{j=1}^k \varphi_j^{pz} z_j^p - \frac{\rho^{xx}}{\varphi^{px}} \left(B - \frac{\varphi^{pp} E}{\Theta_t^r - \varphi^{pp} g^{yy}} + \frac{(-1 - \Theta_t^r + \varphi^{pp} g^{yy}) \left((A + BZ) \left(-\frac{\Theta_t^r}{\varphi^{pp}} + g^{yy} \right) + EZ \right)}{(\Theta_t^r - \varphi^{pp} g^{yy}) D} \right),$$

$$y^p = -\frac{A + BZ + (Z\varphi^{pp} + \varphi^{px2})E}{\varphi^{pp} D},$$

$$p^p = -\frac{(A + BZ) \left(-\frac{\Theta_t^r}{\varphi^{pp}} + g^{yy} \right) + EZ}{\varphi^{pp} D},$$

where $A = \varphi^{px} (-\sum_{i=1}^h \rho_i^{zx} z_i^f - \rho^{tx} t^f + \varphi^x + \sum_{j=1}^k \varphi_j^{xz} (z_j^p - 1) + \varphi^{xt} t^p)$, $B = \varphi^p + \sum_{j=1}^k \varphi_j^{pz} z_j^p + \varphi^{pt} t^p$, $E = g^y + \sum_{k=1}^m g_k^{zy} z_k^f + g^{ty} t^r$, $D = \frac{-Z}{\varphi^{pp}} + (Z + \frac{\varphi^{px2}}{\varphi^{pp}}) (-\frac{\Theta_t^r}{\varphi^{pp}} + g^{yy})$, and $Z = (1 + \Theta_t^p) \rho^{xx} - \varphi^{xx}$.

(A7) Explicit form of pricing Equation (9):

$$p^f = \underbrace{\left(\frac{\varphi^{px} \rho^{xx}}{\rho^{xx} - \varphi^{xx}} + \frac{\varphi^{px}}{\omega_c^{pp}} \right) \lambda_t^p}_{\beta_t^{pp}(\Theta_t^p)} p^p + \underbrace{\left(\frac{-\rho^{zx} \varphi^{xx}}{\rho^{xx} - \varphi^{xx}} + (-\rho^{zx}) \lambda_t^p \right) z^f}_{\beta_t^{zf}(\Theta_t^p)} z^f + \underbrace{\left(\frac{\varphi^{xz} \rho^{xx}}{\rho^{xx} - \varphi^{xx}} + \frac{\varphi^{xz}}{\omega_c^{zp}} \right) \lambda_t^p}_{\beta_t^{zp}(\Theta_t^p)} z^p$$

$$+ \underbrace{\left(\frac{\varphi^{xt} \rho^{xx} - \rho^{tx} \varphi^{xx}}{\rho^{xx} - \varphi^{xx}} + (\varphi^{xt} - \rho^{tx}) \lambda_t^p \right) t^{f+p}}_{\beta_t^{tf+p}(\Theta_t^p)} t^{f+p} + \underbrace{\left(\frac{\varphi^x \rho^{xx} - \rho^x \varphi^{xx}}{\rho^{xx} - \varphi^{xx}} + (\varphi^x - \rho^x) \lambda_t^p \right)}_{\beta_t^c(\Theta_t^p)},$$

$$\text{where } \lambda_t^p = \frac{1}{1 - \frac{\varphi^{xx}}{\rho^{xx}}} \frac{-\Theta_t^p}{1 + \Theta_t^p - \frac{\varphi^{xx}}{\rho^{xx}}}.$$

(A8) Equation of processors' average input conjectural elasticity Θ^p :

$$\Theta_t^p = -\lambda_t^p \frac{\omega_c^{pp}}{\omega_c^p} \left/ \left(\lambda_t^p + \frac{\omega_c^{pp}}{\omega_c^p} \right) \right.$$

(A9) Explicit form of pricing equation:

$$\begin{aligned}
 p^f = & \underbrace{\left(\frac{\rho^{xx} - \rho^{xx} \varphi^{pp} g^{yy}}{\varphi^{px} g^{yy}} + \lambda_t^r \right)}_{\delta_c^{pp}} p^p + \left(\frac{-\varphi^{pz} \rho^{xx}}{\varphi^{px}} \right) z^p + \rho^{zx} z^f + \underbrace{\left(\frac{-g^{zy} \rho^{xx}}{\varphi^{px} g^{yy}} + \underbrace{g^{zy}}_{\delta_\lambda^{z^r}} \lambda_t^r \right)}_{\delta_c^{z^r}} z^r \\
 & + \underbrace{\left(\frac{-(g^{ty} + \varphi^{pt} g^{yy}) \rho^{xx} + \rho^{tx} \varphi^{px} g^{yy}}{\varphi^{px} g^{yy}} + \underbrace{g^{ty}}_{\delta_\lambda^t} \lambda_t^r \right)}_{\delta_c^{t^{f+p+r}}} t^{f+p+r} + \underbrace{\left(\frac{-(g^y + \varphi^p g^{yy}) \rho^{xx} + \rho^x \varphi^{px} g^{yy}}{\varphi^{px} g^{yy}} + \underbrace{g^y}_{\delta_\lambda^c} \lambda_t^r \right)}_{\delta_c^c}
 \end{aligned}$$

where $\lambda_t^r = \frac{-\rho^{xx}}{\varphi^{px} g^{yy}} \frac{\Theta_t^r}{\Theta_t^r - \varphi^{pp} g^{yy}}$.

(A10) Equation of retailers' average input conjectural elasticity Θ^r :

$$\Theta_t^r = -\lambda_t^r \left(\frac{\delta_c^{pp} \delta_\lambda^{z^r}}{\delta_c^{z^r}} + 1 \right) / \left(\frac{\delta_c^{z^r}}{\delta_\lambda^{z^r}} - \lambda_t^r \right).$$